

# Customer Segmentation & Behavioral Analysis Using Machine Learning

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# 1. Introduction

Retail organizations generate large volumes of customer transaction data through daily operations. However, transforming this raw data into actionable insights remains a significant challenge. Customer behavior varies widely in terms of purchasing frequency, spending capacity, and engagement levels. Treating all customers in the same way often leads to inefficient marketing strategies and lower customer retention.

This project applies machine learning techniques to segment retail customers based on their purchasing behavior and feedback. By identifying distinct customer groups, businesses can design targeted marketing strategies, improve customer satisfaction, and optimize revenue generation.

## 1.1 Objectives

- To clean and preprocess a large retail transaction dataset
- To analyze customer behavior using exploratory data analysis (EDA)
- To engineer meaningful features for behavioral analysis
- To apply K-Means clustering to segment customers
- To interpret customer segments from a business perspective
- To present insights through visualizations and dashboards

## 2. Dataset Description

The dataset used in this project is a large-scale retail transaction dataset consisting of approximately 302,000 records. It contains both numerical and categorical variables related to customer demographics, transaction details, product information, and customer feedback.

Source – Kaggle

Link - <https://www.kaggle.com/datasets/bhavikjikadara/retail-transactional-dataset>

## 2.1 Key Attributes

- Customer information: Age, Gender, Country, City
- Transaction data: Total\_Purchases, Amount, Total\_Amount, Date, Time
- Product data: Product\_Category, Product\_Brand, Product\_Type
- Customer feedback: Ratings, Feedback

The dataset includes missing values, duplicate records, and outliers, making preprocessing a critical step before model development.

## 3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the structure and characteristics of the dataset.

### 3.1 Summary Statistics

- Customer ages range from 18 to 70 years
- Ratings range from 1 to 5, with most values concentrated around 3 to 4
- Total purchases and Total amounts show a wide variation, indicating diverse spending behavior

### 3.2 Missing Value Analysis

Several columns contained missing values, though the percentage of missing data per column was relatively low (generally below 1%). A structured approach was adopted to handle these missing values appropriately.

### 3.3 Outlier Detection

Boxplots were used to identify outliers in numerical variables such as Total\_Purchases and Total\_Amount. The observed extreme values are particularly in spending-related features, which could bias clustering results if left untreated.

## 4. Data Preprocessing

To ensure data quality and reliability, the following preprocessing steps were performed:

### Handling Missing Values

- Categorical variables were filled with the label "Not Provided."
- Numerical variables were imputed using mean or median values, depending on their distribution.

### Data Type Corrections

- Identifier fields such as Customer\_ID and Transaction\_ID were converted to string format.
- Age was converted to an integer data type.

### Duplicate Removal

Duplicate records were identified and removed to prevent skewed analysis.

### Outlier Treatment

Outliers in Total\_Purchases and Total\_Amount were handled using a quantile-based approach, removing values above the 99th percentile.

## 5. Feature Engineering

Feature engineering was conducted to enhance the dataset for clustering analysis.

### Date-Time Processing

- The Date column was converted to a datetime format
- Year and Month features were extracted

### **Derived Features**

- Average Order Value: Total\_Amount divided by Total\_Purchases
- Spend per Purchase: Captures spending intensity per transaction

These engineered features provide deeper insights into customer purchasing behavior beyond raw transaction counts.

## **6. Machine Learning Model Development**

### **Model Selection**

This project uses **K-Means Clustering**, an unsupervised learning algorithm suitable for identifying natural groupings within data.

### **Feature Selection**

The following numerical features were selected for clustering:

- Age
- Total\_Purchases
- Total\_Amount
- Avg\_Order\_Value
- Ratings

### **Data Scaling**

StandardScaler was applied to normalize the data, ensuring that features with larger magnitudes did not dominate the clustering process.

### **Determining Optimal Number of Clusters**

The Elbow Method was used to identify the optimal number of clusters. Based on the inertia plot, four clusters were selected as they provided a balance between compactness and interpretability.

## 7. Customer Segmentation Results

After applying K-Means clustering with four clusters, customers were segmented into the following groups:

### **Cluster 0: Regular Customers:**

- Moderate purchase frequency
- Moderate total spending
- Average ratings
- Represent stable customers with consistent purchasing behavior

### **Cluster 1: High-Value Loyal Customers:**

- High purchase frequency
- Highest total spending
- High average order value
- Strong contributor to overall revenue

### **Cluster 2: At-Risk Customers:**

- Low purchase frequency
- Low total spending
- Lowest ratings
- High likelihood of churn

### **Cluster 3: Premium Occasional Buyers:**

- Low purchase frequency
- High spending per purchase
- Highest ratings
- Purchase infrequently but spend significantly per transaction

PCA was used to reduce dimensionality and visualize the clusters effectively.

## 8. Dashboard and Visualization

To effectively communicate the results of the customer segmentation model, an interactive dashboard was developed using **Power BI Desktop**. The dashboard integrates machine learning outputs with business performance metrics, enabling both high-level monitoring and detailed diagnostic analysis.

The dashboard is organized into two main pages: **Business Overview** and **Diagnostics Analysis**, allowing users to explore customer behavior across different segments and time periods.

### 8.1 Key Visuals

#### 1. Customer Distribution by Segment

This visualization presents the proportion of customers in each cluster identified by the K-Means model. The distribution indicates a relatively balanced segmentation, ensuring that no single cluster disproportionately dominates the dataset.

##### **Insight:**

The presence of sizable customer groups across all clusters highlights the need for differentiated marketing strategies rather than **a one-size-fits-all approach**.

#### 2. Revenue Contribution by Cluster

This bar chart illustrates total sales generated by each customer segment.

##### **Insight:**

The **High-Value Loyal Customers** cluster contributes the highest share of total revenue despite not being the largest segment. This confirms that a smaller group of loyal customers drives a significant portion of overall sales.

#### 3. Purchase Frequency by Cluster

This visualization compares the average number of purchases across customer segments.



**Insight:**

High-value loyal customers exhibit the highest purchase frequency, while **At-Risk Customers** and **Premium Occasional Buyers** show lower engagement levels. This highlights differences in loyalty and repeat behavior among segments.

**4. Customer Satisfaction by Ratings and Product Category**

This stacked bar chart displays customer ratings across product categories.

**Insight:**

Higher ratings are more common among Premium Occasional Buyers, suggesting that although they purchase infrequently, they experience high satisfaction. Lower ratings observed in the At-Risk segment indicate potential dissatisfaction issues.

**5. Repeat Customers Over Time**

This line chart tracks repeat customer trends over monthly periods.

**Insight:**

Fluctuations in repeat purchases indicate seasonal or behavioral changes. A noticeable decline in the last few months for certain segments signals potential churn risks that require proactive intervention.

**Drill Through Analysis**

The dashboard incorporates a drill-through analysis focused on customer clusters, allowing users to examine detailed performance insights for each segment. By selecting a specific customer cluster, users can view transaction trends and key metrics related to that segment. This feature supports deeper understanding of customer behavior across different clusters and enables management to design more targeted retention

## 9. Challenges

Several challenges were encountered during the development of this self-project:

- **Data Quality Issues:** Several actions were taken to adjust missing values, inconsistent data types, and outliers, which required extensive preprocessing.
- **Feature Selection Complexity:** Identifying the most relevant features for clustering involved multiple iterations and validation.
- **Cluster Interpretability:** Translating numerical clusters into meaningful business segments required considerable time and careful analysis.
- **Integration Complexity:** Aligning machine learning outputs with Power BI dashboards required schema adjustments and data modeling refinements.

## 10. Business Recommendations

Based on the clustering results and dashboard insights, the following business recommendations are proposed:

### High-Value Loyal Customers

- Implement loyalty reward programs and exclusive offers.
- Focus on personalized communication to retain this high-revenue segment.

### Regular Customers

- Encourage higher purchase frequency through targeted discounts.
- Introduce bundle offers to increase basket size.

### Premium Occasional Buyers

- Promote limited-time premium offers to stimulate repeat purchases.
- Use personalized reminders and product recommendations.

## **At-Risk Customers**

- Introduce retention campaigns such as discount vouchers or re-engagement emails.
- Address dissatisfaction through improved customer service and feedback analysis.

## **11. Conclusion**

This project successfully demonstrated how machine learning can be used to extract actionable insights from large-scale retail data. Through comprehensive preprocessing, feature engineering, and K-Means clustering, customers were segmented into meaningful behavioral groups. The integration of clustering results into an interactive Power BI dashboard enabled clear visualization of customer behavior, revenue contribution, and engagement patterns.

The findings highlight the importance of customer-centric strategies and show how data-driven segmentation can significantly enhance marketing effectiveness, customer retention, and revenue optimization.

## **12. Future Enhancements**

Several improvements can be considered to extend this project further:

- Incorporating **RFM analysis** for enhanced behavioral segmentation.
- Including **real-time data updates** for dynamic dashboard monitoring.
- Developing **predictive churn models** using supervised learning.